

MS 491 : MARKETING ANALYTICS · GROUP ASSIGNMENT 1



Four Markets, Four Constraints

What 90,161 Airbnb listings in Belgium, Netherlands, Rio de Janeiro and Los Angeles reveal about travellers, hosts and the platform - and what Airbnb should do about it?

90,161

listings analysed

62,702

distinct hosts

4

markets, 3 currencies

3

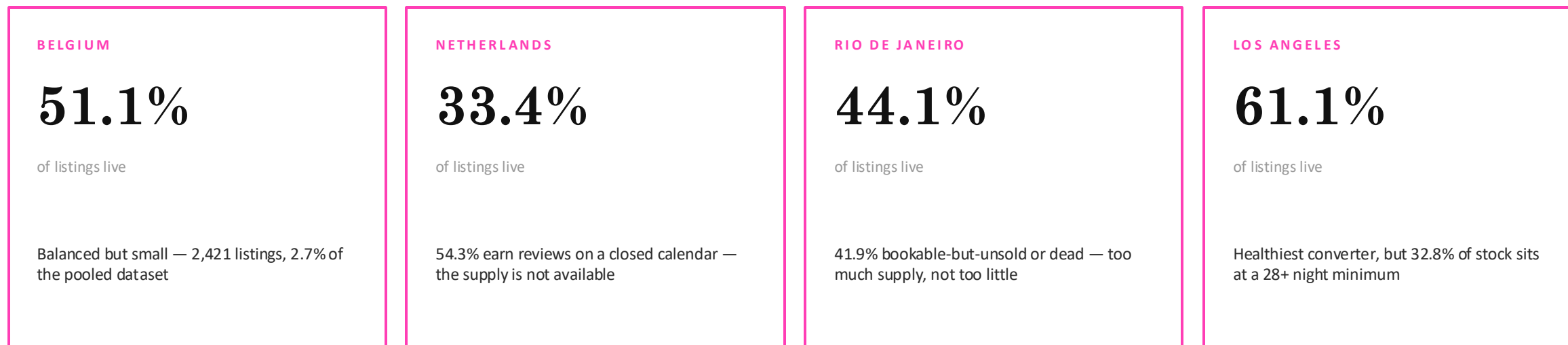
stakeholder perspectives

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Listing count is the wrong measure of market size — and it is wrong in a different way in every market

Only 33–61% of listings are both bookable and actually transacting. The gap isn't noise - it's closed calendars in Amsterdam, unsold inventory in Rio, and long-stay stock in LA. Budgeting on listing count gets three of four markets wrong



The strategic question is not where to add supply. It is which constraint is binding — and whether Airbnb can move it.





Four markets, one schema, three currencies — comparability had to be engineered, not assumed

90,161

listings

62,702

distinct hosts

499

neighbourhoods

16

columns per listing

Market	Listings	Hosts	Neighb.	Median price	Max price
Antwerp · Belgium	2,421	1,568	57	€70	€4,600
Amsterdam · Netherlands	18,949	16,410	22	€130	€8,000
Rio de Janeiro	35,724	24,992	156	R\$300	R\$132,358
Los Angeles · California	33,067	19,741	264	\$109	\$24,999

HOW THE MARKETS WERE MADE COMPARABLE

- Price index — Each listing divided by its market’s own median price. No currency conversion needed.
- Compared within neighbourhood x room type – pooled comparisons can flip sign (Simpson’s paradox)
- Reviews per month tracks current demand, total reviews just tracks age

WHAT THE POOLED MARKET LOOKS LIKE

- Entire homes are 70% of all listings, but the mix varies sharply — Los Angeles is the most room-diverse market at roughly a third private rooms.
- Host concentration ranges from 21% of listings (Amsterdam) to 54% (Los Angeles) belonging to a host running two or more properties.
- Review activity spans Mar 2010 – Aug 2020, placing the snapshot around mid-2020.

CONTEXT

Rio and Los Angeles hold 76% of the pooled listings, so any statistic computed across all four is dominated by two markets and describes none of them. Every finding in this deck is computed market by market.



The same number means three different things depending on who is asking — so we ask three times

THE TRAVELLER

deciding where to go and what to book

What does my budget actually buy — a whole place or somebody's spare room?

Does booking longer really save money?

Can I actually book this, or is it decoration in the search results?

THE HOST

deciding whether and how to list a property

Where do I fit — and what should I charge?

Which levers actually get me booked?

Does the capital math work, and what could take it away?

THE REGIONAL MANAGER

allocating marketing budget across markets

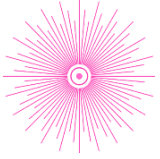
Which market's supply is not converting into bookings — and why?

Where is demand outrunning supply, neighbourhood by neighbourhood?

Is this market's price data describing real sales, or unsold inventory?

WHY IT MATTERS

The three perspectives are not three separate assignments. Traveller behaviour drives host incentives, host incentives shape what the platform can sell, and regional conditions decide which strategy is even available. The final three slides put them back together.



PERSPECTIVE ONE

The Traveller

Three questions a traveller asks before booking. Each is tested market by market, with budget expressed as a percentile of the market's own prices so no currency conversion is needed.

Q1

What does an ordinary budget actually buy — a whole place, or somebody's spare room?

Q2

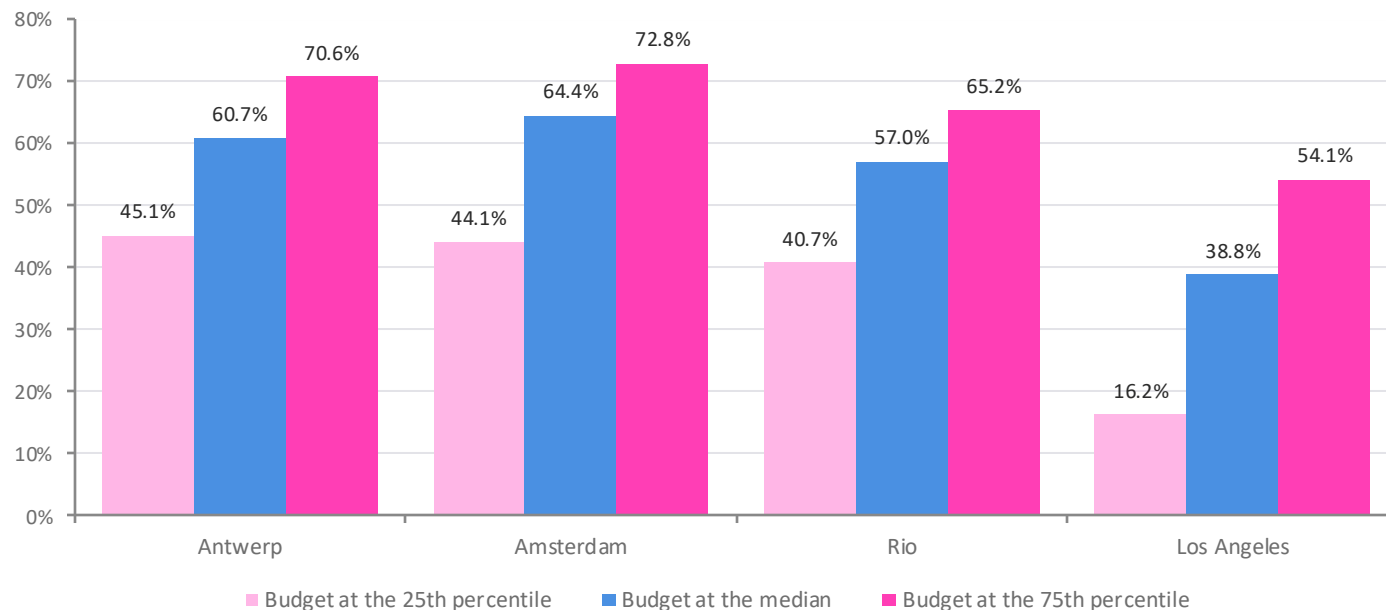
Does committing to a longer stay reliably save money, everywhere?

Q3

Can you actually book this, or is it decoration in the search results?

The same relative budget buys a door you can close in Amsterdam and a shared corridor in Los Angeles

THE QUESTION *"Can I afford a door I can close?"*



BUDGET PERCENTILE FOR A 50/50 SHOT AT AN ENTIRE HOME

P28

Amsterdam

P34

Antwerp

P37

Rio

P66

Los Angeles

A 38-percentile-point spread on the most basic traveller decision.

VALIDATING THE PREMISE

Entire homes cost more than private rooms in 316 of 320 neighbourhood tests (98.8%) — 100% in Antwerp, Amsterdam and Los Angeles, 95.2% in Rio. The direction is effectively a law; only the composition of the stock varies.

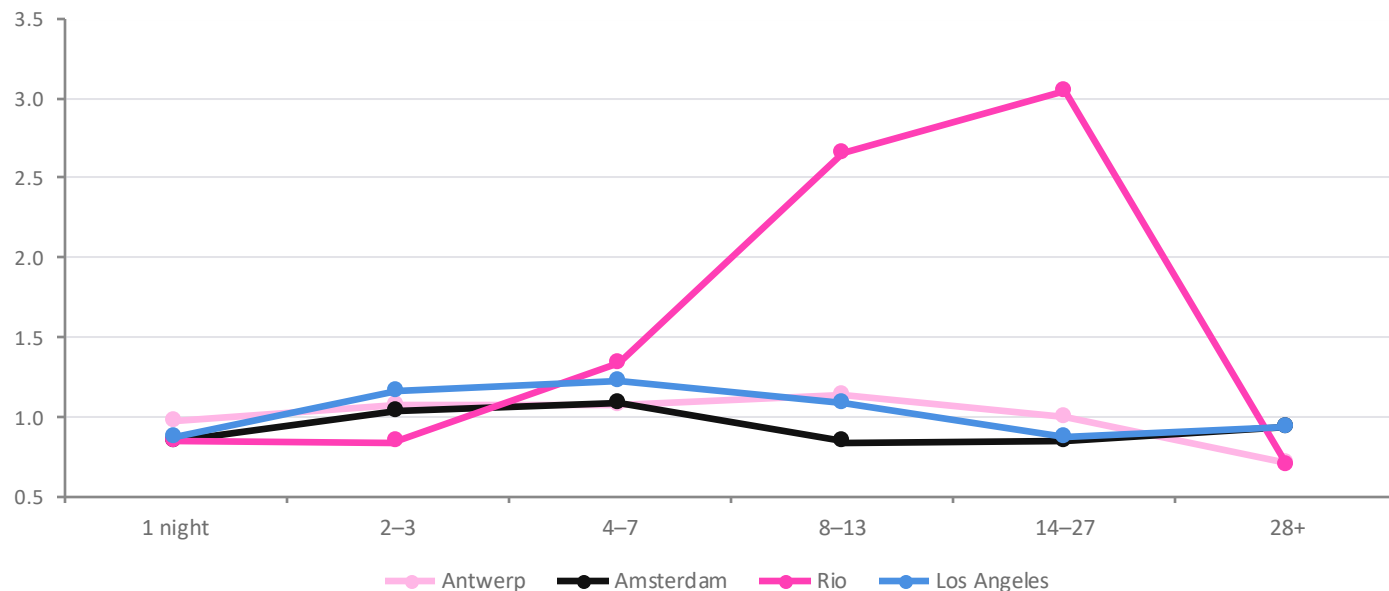
Share of the stock a traveller can afford that is an entire home. Budget = everything at or under that percentile of the market's own price distribution.

IMPLICATION

"An entire home for less than you'd expect" is a genuinely differentiated claim in Los Angeles, where it takes a top-third budget. In Amsterdam it is already the median experience, so the same copy sells nothing — there the differentiated claim has to be location or format, not privacy.

“Book longer and you'll save” survives an endpoint test and fails a curve test — in the two largest markets

THE QUESTION “If I book two weeks instead of two nights, do I get a discount?” The instinctive answer is yes.



Median price index by minimum-nights bucket. 1.0 = that market's own median nightly price. Rio's 14-27-night bucket sits at 3.05x — the middle of the range is the expensive part, not the cheap one.

DOES PRICE VARY AT ALL ACROSS BUCKETS?

Price does vary by length of stay in all four markets (Kruskal-Wallis, all $p < 0.001$). But that just says it's not flat — not whether it's a gentle slope or a sharp hump.

DOES THE RELATIONSHIP BEND?

It's a hump: Rio (+0.354) and LA (−0.232) both curve significantly ($p < 0.001$). Antwerp and Amsterdam don't — just a flat, unremarkable relationship.

TWO ROBUSTNESS CHECKS

Comparing just the two endpoints (short vs. long), the ratio favors longer stays everywhere (1.25 / 1.06 / 1.30 / 1.22) — which makes the naive rule look true. It isn't: capping outlier prices in Rio doesn't change the hump, so it's not just a few luxury listings skewing things.

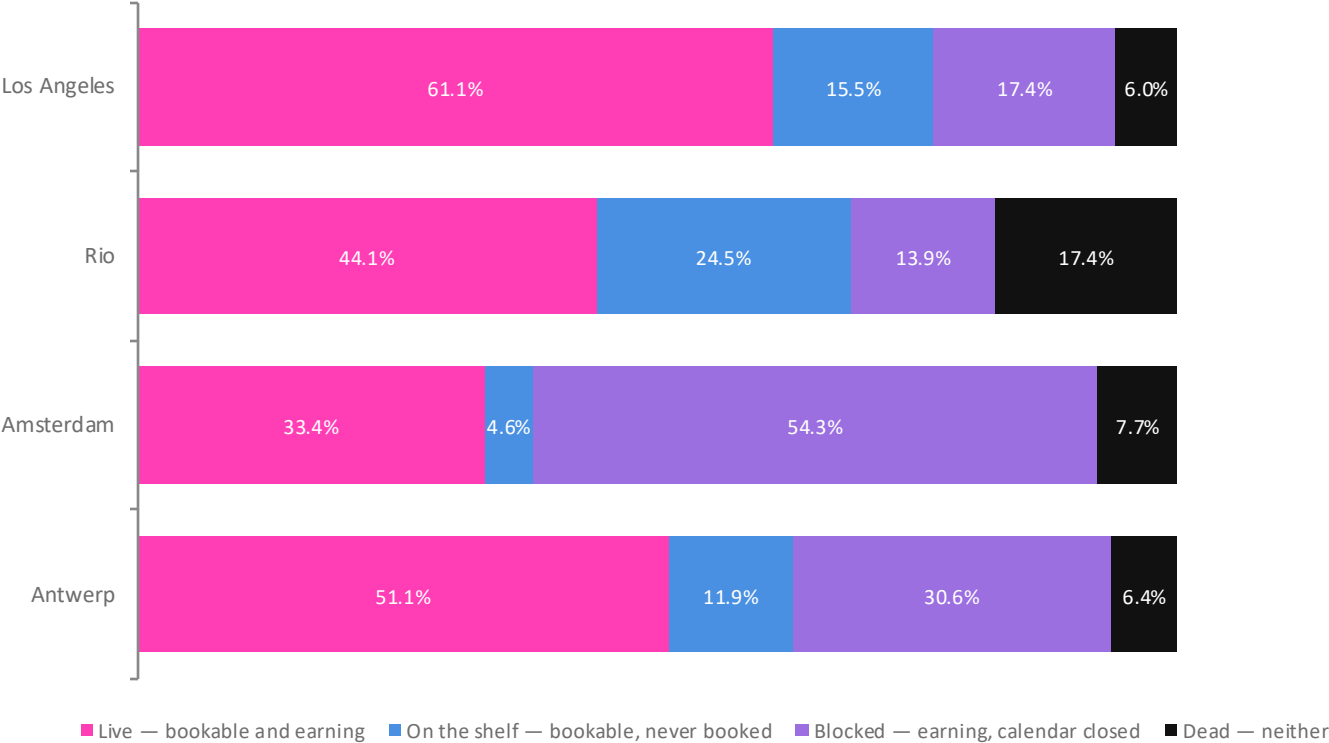
IMPLICATION

A discount claim needs to name its range — the extremes-only comparison looks universal but fails for the mid-length stays people actually book in Rio and LA. LA's also different: a third of its listings sit at 28+ nights, out of the short-stay race entirely.

A third to two-thirds of what a traveller sees is not really for sale — and the risk concentrates at the top of the price range



THE QUESTION “Can I actually book this, or is it decoration in the search results?” A listing can stay active while its calendar is effectively closed.



THE MIX IS GENUINELY DIFFERENT
Chi-square test of independence between inventory health and market: $\chi^2 = 16,882.2$, $df = 9$, $p \approx 0$.

BLOCKED: REGULATION OR POPULARITY?
Amsterdam's blocked share is nearly double for entire homes vs. private rooms, everywhere else the gap is ~3 points

Share of listings that have never been reviewed

	Whole market	Dearest quartile
Antwerp	18.2%	29.6%
Amsterdam	12.3%	15.6%
Rio de Janeiro	42.0%	71.2%
Los Angeles	21.5%	28.3%

IMPLICATION

A review-count filter is not a nicety in Rio — it is the difference between browsing inventory and browsing sales, where seven of ten of the most expensive listings have never been booked. Amsterdam is the mirror image: almost everything bookable there has a track record.

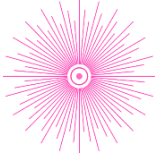


No two markets tell the same story on all three questions — which is why one traveller message cannot work across four

	BELGIUM	NETHERLANDS	RIO DE JANEIRO	LOS ANGELES
QUESTION 1				
Budget needed for a 50/50 shot at an entire home	Middke of the pack – a flat market.	Easiest of the four.	Middle, but a huge price ladder.	The outlier – needs the biggest budget.
QUESTION 2				
Does a longer stay reliably save money?	No clear trend. No significant curvature.	No clear trend. A flat, boring market.	No — mid-length stays cost most.	No — same hump shape, plus a third of stock needs 28+ nights.
QUESTION 3				
Headline bookability failure	None dominant – likely just small.	Regulatory – calendars closed.	Oversupply – inventory does not sell.	Healthiest, but priced for monthly stays.

SECTION TAKEAWAY

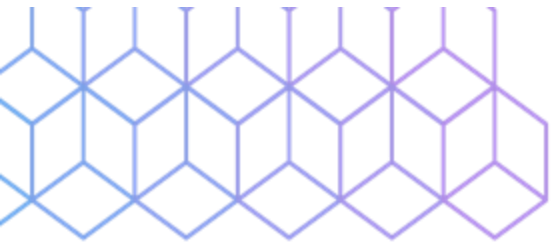
Rio and Los Angeles behave like each other on Question 2 — the same statistically real price hump — but like opposites on Questions 1 and 3: Rio is budget-friendly with an oversupply problem, Los Angeles is expensive to reach a whole place in but has the healthiest-converting inventory. A single cross-market pricing or marketing message would be wrong somewhere on every question.



PERSPECTIVE TWO

The Host

Four decisions in the order a real host makes them: where do I fit → how do I run it → does the money work → what could take it away.



Q1

Where do I actually fit in this market?

Q2

How do I configure the listing to get booked?

Q3

Does the capital math actually work?

Q4

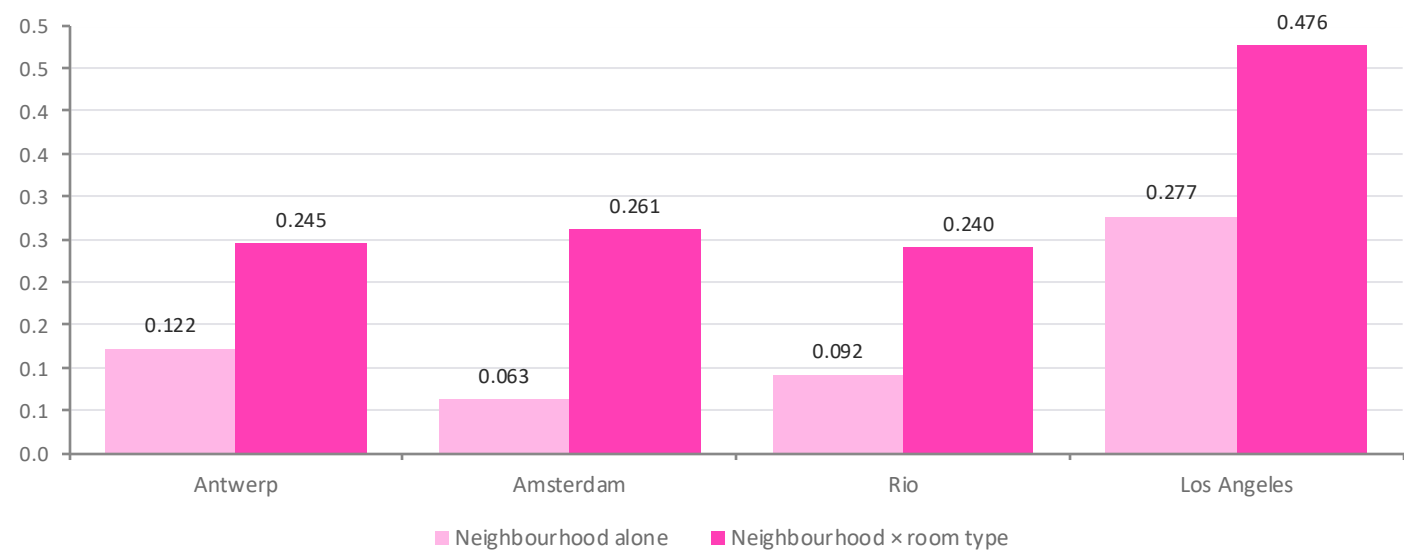
What could take this away from me — and is the income even real?



Comparable listings pin down half of a price in Los Angeles and less than a quarter of one in Rio

THE QUESTION “Where do I fit — what is normal here, and is guessing the citywide average safe or dangerous?”

Variance of log price explained by neighbourhood × room type



A low R^2 is not a weak analysis — it means price is not determined by the variables we have, which is itself useful to a host.

LOW R^2 DOES NOT ALWAYS MEAN THE SAME THING

	Comps explain	Spread still unexplained
Antwerp	24.5%	1.98×
Amsterdam	26.1%	1.59×
Rio de Janeiro	24.0%	2.59×
Los Angeles	47.6%	1.91×

THE COST OF A LAZY GUESS

Guessing the citywide median instead of pricing your own neighbourhood is off by 43% in Rio — worst of the four. Its price spread is also roughly double Amsterdam's or Antwerp's, so it's the market that punishes shortcuts hardest.

WHO YOU ARE COMPETING WITH

Professional hosts (3+ listings) make up 39.2% of LA and 33.4% of Antwerp, but just 9.7% of Amsterdam. Antwerp's professionals are a few large operators; LA's are many mid-sized ones. A new host in Amsterdam is joining a market of amateurs.

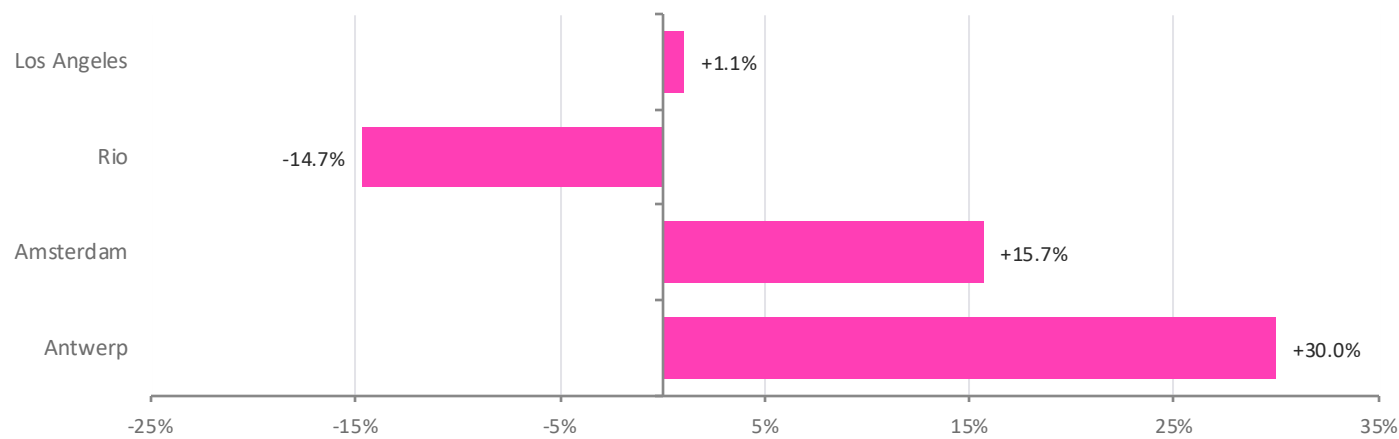
IMPLICATION

Airbnb's pricing guidance should be calibrated to its own reliability rather than presented with uniform confidence. In Los Angeles, show the comps. In Rio, show the comps and the spread — two otherwise-identical listings in the same neighbourhood and format still differ by a factor of 2.6.

Two levers behave the same everywhere and two reverse outright — the same instinct is right in Antwerp and wrong in Rio

THE QUESTION “What should I actually do — what price, what minimum stay, what calendar policy — and how confident should I be?”

Price premium of professional (3+ listing) hosts, within neighbourhood × room type



CELLS WHERE PROFESSIONALS SELL MORE

Antwerp 31% Amsterdam 55% Rio 71% Los Angeles 68%

Antwerp's professionals charge 30% more and sell less; LA's charge the same and sell more — competence, not pricing power. Pooled across markets the pattern reverses (Simpson's paradox), which is why we read this within-market only.

SAME EVERYWHERE · MINIMUM NIGHTS

A longer minimum stay costs bookings everywhere — in 92–96% of cells across all four markets. Steepest in LA: going from a 1-night to a 4–7-night minimum cuts booking rate by roughly two-thirds.

SAME EVERYWHERE · WHOLE-PLACE PREMIUM

Entire homes command 1.7× to 2.7× a private room in the same neighbourhood. Direction universal, magnitude varies.

REVERSES · OPENING THE CALENDAR

Weakly positive in Antwerp (+0.028) and Amsterdam (+0.061); negative in Rio (−0.166) and LA (−0.085). Likely reverse causation — 300 open days usually means unsold, not available.

REVERSES · THE REVIEW-COUNT TRAP

More accumulated reviews correlates with lower price, steepest in Rio — but that's reverse causation: cheap listings book faster and rack up reviews faster. A data warning, not a pricing tip.

IMPLICATION

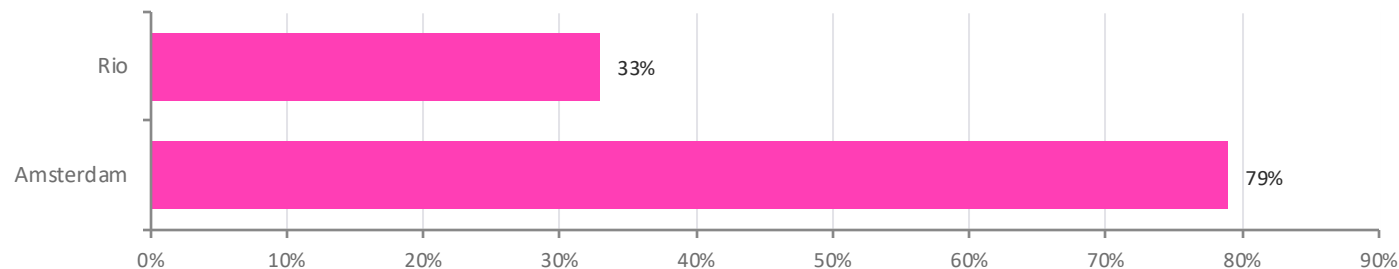
Two rules travel globally: shorten your minimum stay, price the whole place at a premium. Two don't — “open your calendar” is wrong in Rio, and “price like a professional” is right in Antwerp but a 15% mistake in Rio.



Buying a property to run as an Airbnb business in Amsterdam is not weak economics — it is arithmetically blocked

THE QUESTION “Should I buy something to run this as a business?” The first host question that needs data from outside the listings file.

Occupancy needed for a 10-year payback, against what the law allows



WHAT EACH MARKET'S LAW PERMITS, AS A SHARE OF THE YEAR

Antwerp
no cap

Amsterdam
30 nights = 8%

Rio
no cap

Los Angeles
120 nights = 33%

WHAT THIS CALCULATION ASSUMES

Property cost per m² (€3,100 Antwerp, €8,500 Amsterdam, \$6,458 LA, R\$9,500 Rio), a 50m² unit, 10-year payback, and each market's median entire-home price. FX just puts the four on one axis.

WHY THIS IS SCENARIO ARITHMETIC, NOT A FINDING

None of these cost, wage, or FX inputs come from the listings data — this is scenario math, not a measured result. But the Amsterdam gap is too large for reasonable assumption changes to close.

THE SINGLE SHARPEST NUMBER IN THE HOST ANALYSIS

Amsterdam needs ~79% occupancy to break even in 10 years, but its own law caps unhosted rentals at ~8%. The legal ceiling sits at a tenth of the required floor. Rio's the opposite: cheapest breakeven at 33%, and the best pricing power relative to property cost (30% vs. Amsterdam's 13%).

WHAT IT IS USEFUL FOR

Ranking, not forecasting — it shows which markets are worth modelling properly, and where acquisition spend can't convert no matter how good the marketing is.

IMPLICATION

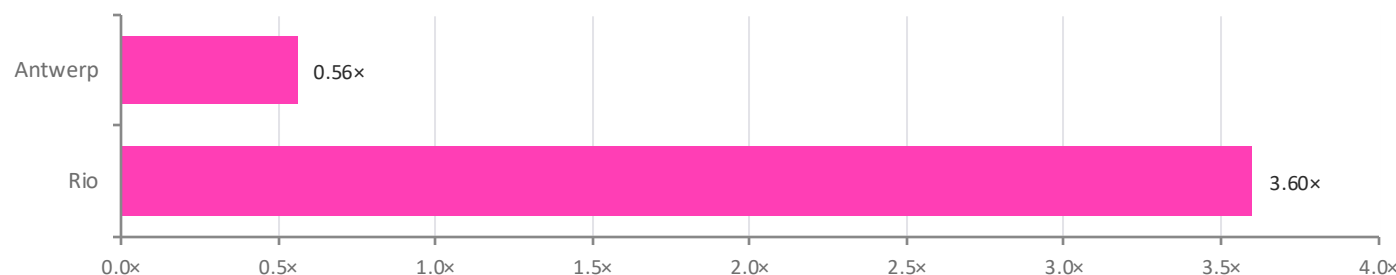
Amsterdam's constraint is statutory, not commercial — no price recruits investor-hosts there, so budget should go to the one format the cap doesn't touch. Rio has the strongest capital case, which is exactly why its oversupply keeps getting worse.



The markets where Airbnb pays best relative to local wages are the same markets regulators are moving against first

THE QUESTION *“What could take this away from me — and is the income even real?”*

15 booked nights a month at the local median entire-home price, as a multiple of local minimum wage



THE SAME ACTIVITY, TWO DIFFERENT ECONOMIC OBJECTS

In Rio, a 15-night month nets ~3.6x local minimum wage — investment-grade income, not a side hustle. In Antwerp the same calendar earns 0.56x: just extra cash. That gap is why Rio faces condo votes and proposed bans while Antwerp doesn't.

THE LIMIT ON ALL OF THIS

Regulation here is inferred from patterns, not statute — the 30-night spike is consistent with LA's rental ordinance, but the data can't prove causation. “Consistent with,” never “caused by.”

REGULATORS ARE ALREADY WHERE THE MONEY IS

The neighbourhoods targeted for 2026 restrictions — Amsterdam Centrum and Rio's Zona Sul — are both more professionalised than their citywide averages. The places earning the most relative to local wages are moved against first.

WHERE THE EXPOSURE ALREADY SHOWS IN THE DATA

LA is the only market with a visible regulatory fingerprint: 30.2% of listings sit at exactly 30–31 nights — a spike that can't occur naturally, vs. 1% or less elsewhere. Amsterdam's constraint shows up differently: 54.3% blocked calendars instead.

IMPLICATION

Regulatory exposure is the binding constraint in two of the four markets, not a footnote. Airbnb's best options: compete on formats regulation doesn't touch and treat the income multiple as an early-warning signal for the next restriction.

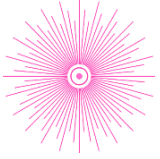


No market wins on all four questions — the right host strategy is different in each, and opposite in two

	BELGIUM	NETHERLANDS	RIO DE JANEIRO	LOS ANGELES
QUESTION 1				
Where do I fit?	Comps explain little. A third of the market is professional and concentrated.	Tight spread – little variation to price into. Mostly amateurs.	Widest spread. A citywide price guess is 43% off.	Comps explain 47.6% — by far the most reliable.
QUESTION 2				
How do I configure it?	Pricing like a professional pays more but they sell less.	Pricing edge, but few professionals to model.	Pricing like a professional is a mistake – volume wins here.	Price does not matter – competence does.
QUESTION 3				
Does the money work?	Lowest capital risk, no cap.	Blocked outright by the cap.	The strongest capital case of the four.	Workable, but capped by ordinance.
QUESTION 4				
What could take it away?	Pocket money – low regulatory risk.	Cap already binding; private room exempt.	Investment-grade income – high regulatory target.	Ordinance already reshaping the market.

SECTION
TAKEAWAY

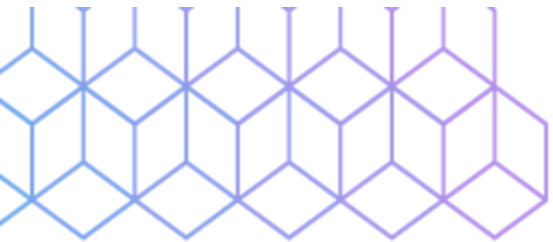
Rio is highest-reward and highest-scrutiny; Antwerp is calmest ,modest income, lowest capital risk; Amsterdam is capital-unfriendly by regulation despite reasonable day-to-day economics; LA is the least extreme overall, and the only market where the comps can be trusted.



PERSPECTIVE THREE

The Regional Manager

Allocating budget across markets. The useful questions distinguish “no supply” from “supply nobody wants” from “supply that cannot legally be sold”.



P2.1

Which market's supply is not converting into bookings — and why not?

P2.2

Where inside each market is demand outrunning supply?

P2.3

Which of these markets is actually cheap?

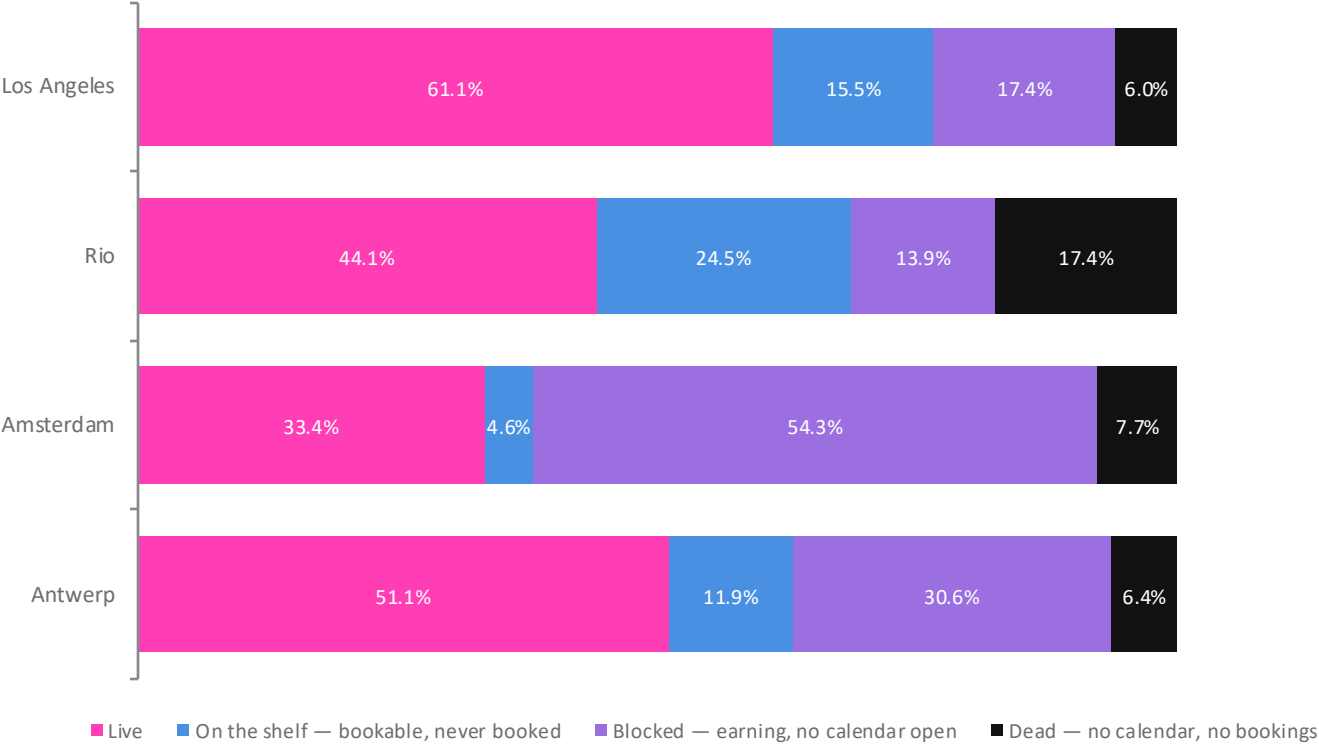
P2.4

Are private rooms an underused asset or dead weight?



The four markets land in four different quadrants — and each failure mode implies a different intervention

THE QUESTION “Which market’s supply is not converting into bookings, and why not?” The core budget question.



AMSTERDAM — A SUPPLY PROBLEM
54.3% of listings earn reviews on a fully closed calendar — what a 30-night cap looks like in the data. Advertising buys nothing here; the constraint is regulatory, not awareness.

RIO — THE OPPOSITE FAILURE
24.5% is bookable but never sold, plus 17.4% neither bookable nor selling. Rio has too much supply, not too little — adding listings there works against the actual problem.

LOS ANGELES AND ANTWERP
LA is healthiest at 61.1% live. Antwerp is middling across the board — likely just because it's the smallest market.

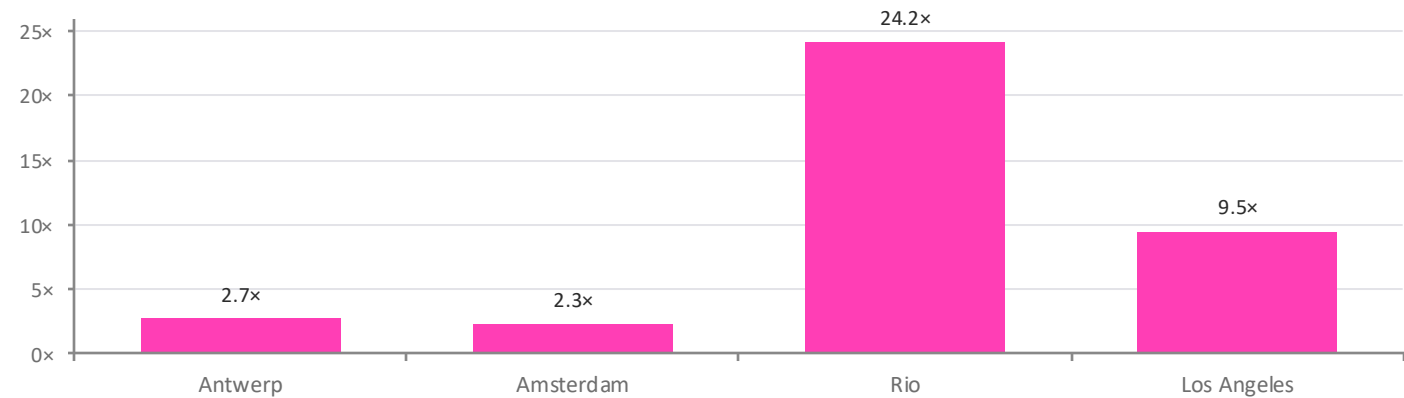
IMPLICATION

Short answer to “put budget behind Rio instead of Belgium”: no. Rio's problem is demand, Antwerp's is just size — neither is fixed by the intervention listing count would suggest.



Rio and Los Angeles hide a 24× and 10× internal imbalance behind a balanced-looking market average

THE QUESTION “Where inside each market is demand outrunning supply?” Media is bought at neighbourhood level, not market level.



RIO DE JANEIRO

The typical Rio neighbourhood earns about half the demand its listing count justifies (median 0.53). Ipanema (1.55) and Copacabana (1.40) absorb it; Engenho de Dentro sits at 0.06. Copacabana alone is 26% of the market's listings — oversupply stacked in places that are already full.

	Testable	Median	10th pct	90th pct
Antwerp	16	0.98	0.57	1.52
Amsterdam	22	1.00	0.77	1.44
Rio de Janeiro	49	0.53	0.25	1.25
Los Angeles	127	0.97	0.62	1.45

LOS ANGELES — THE OBVIOUS ANSWER IS WRONG

The hot spots aren't the famous ones — Del Aire (3.01) and El Segundo (2.32) beat Manhattan Beach (0.32) and Bel-Air (0.37) by 7×. LA's largest neighbourhood is only 5.1% of stock, so the imbalance is spread out, not concentrated in one district.

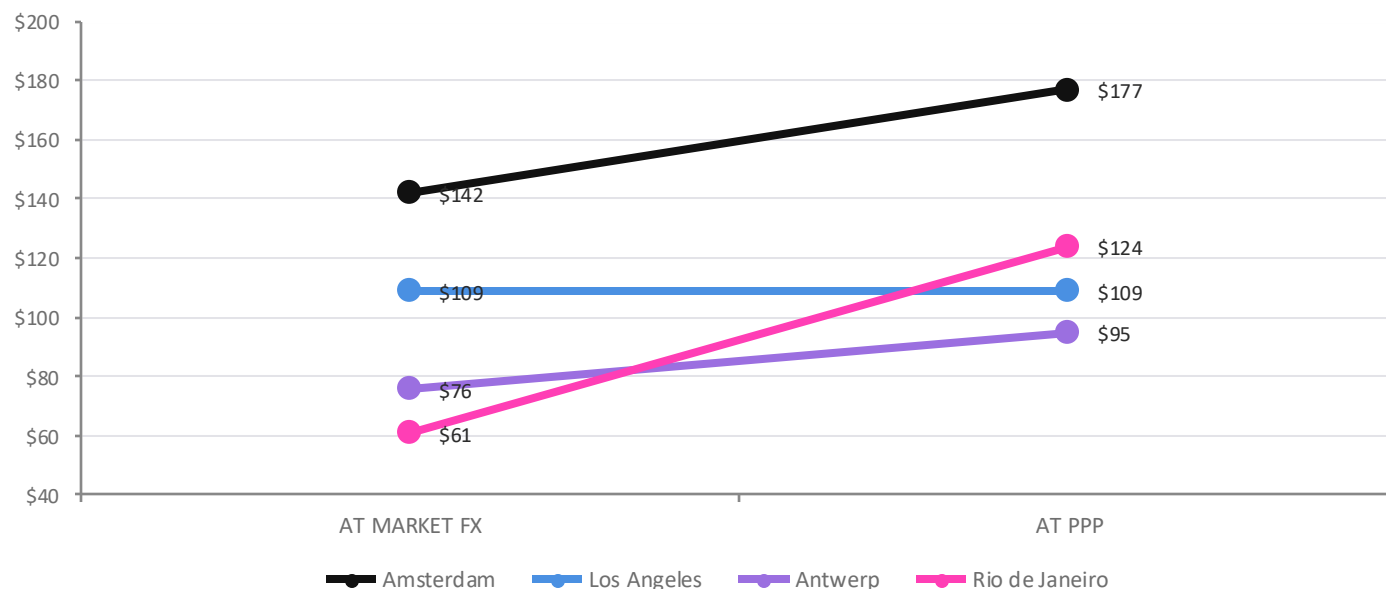
IMPLICATION

Antwerp and Amsterdam are internally efficient — no obvious place to target budget. In Rio and LA, market-level media spend is the wrong resolution: recruit in the 90th-percentile neighbourhoods, pull spend from the bottom decile, and don't assume the prestige neighbourhood is the profitable one.



Rio is the cheapest market to visit and the third-cheapest to live beside — the ranking reverses with the conversion you pick

THE QUESTION “Belgium looks expensive — should I put the budget behind Rio instead?” The honest answer depends on a choice most people make without noticing.



Median nightly price converted to USD. Local medians: €70 Antwerp, €130 Amsterdam, R\$300 Rio, \$109 Los Angeles.

FX ANSWERS THE TRAVELLER'S QUESTION

“What does this cost a visitor with dollars?” The right lens for inbound-tourism marketing — and under it, Rio is the cheapest of the four by a wide margin.

PPP ANSWERS THE HOST'S QUESTION

“What does this cost relative to local incomes?” The right lens for host recruitment. Under it, Rio drops to third of four and Antwerp becomes cheapest.

ONLY ONE RANK IS STABLE

Amsterdam is the dearest market under every normalisation. Everything else moves — Rio and Antwerp swap ends of the table outright. Neither conversion is wrong; they answer different questions.

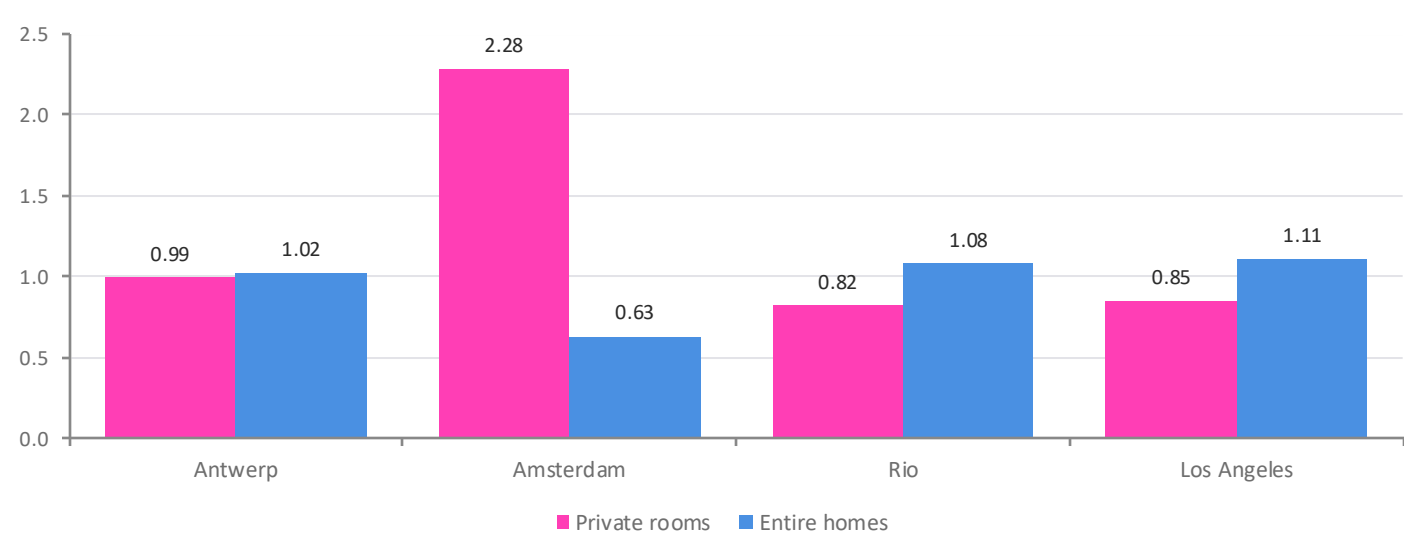
IMPLICATION

P2.1 showed Rio's constraint is demand, not supply — so market it to travellers as the genuinely cheap option it is (FX), not as a cheap place to recruit hosts (PPP), which would only deepen the oversupply. It's also why the rest of the deck uses a price index.



In the one market where Airbnb cannot add whole homes, private rooms already work twice as hard as their share of supply

THE QUESTION “Are private rooms an underused asset or dead weight?” Format is one of the few levers the platform can pull directly.



AMSTERDAM REVERSES THE PATTERN OF THE OTHER THREE

Private rooms are a fifth of Amsterdam's listings but nearly half its bookings — over twice their supply share, while entire homes sit at 0.63. Antwerp's formats are in equilibrium; in Rio and LA, entire homes carry the market.

THE SAME FACT, SEEN TWICE

P2.1 found 54.3% of Amsterdam's listings blocked, concentrated in entire homes. This 2.28 index is that phenomenon from the demand side — the two are one finding.

	Antwerp	Amsterdam	Rio	LA
Private rooms, share of listings	22.7%	20.7%	25.6%	32.6%
Private rooms, share of reviews	22.4%	47.1%	20.9%	27.5%

IMPLICATION

Recruit private rooms in Amsterdam — scarce relative to demand, and the format least constrained by the cap. Skip it in Rio and LA, where private rooms underperform and entire homes carry the market.



Budget should follow convertible supply, not supply — and on that measure the ranking changes in three of four markets

	BELGIUM	NETHERLANDS	RIO DE JANEIRO	LOS ANGELES
P2.1 Is supply converting?	51.1% live, no dominant failure mode — middling on every cut.	33.4% live, 54.3% blocked. A regulatory supply constraint, not a demand gap.	44.1% live, 41.9% bookable-but-unsold or dead. Too much supply, not too little.	61.1% live — the healthiest converter of the four.
P2.2 Where inside the market?	Internally efficient - No obvious place to point a budget.	Internally efficient at 2.3× across 22 neighbourhoods.	Large internal imbalance, concentrated in one district.	Imbalanced too, and the hot spots are not the famous ones.
P2.3 Is it actually cheap?	\$76 at FX, \$95 at PPP — cheapest once local incomes are factored in.	\$142 at FX, \$177 at PPP — dearest under every normalisation. The only stable rank.	\$61 at FX, \$124 at PPP —cheapest to travelers, not to the host.	\$109 either way — the USD anchor, and unmoved by the choice of conversion.
P2.4 Which format to recruit?	Formats in equilibrium. No action.	Recruit private rooms: the cap does not reach them.	Private room underperforming. Entire homes carry the market.	Not a format lever here.

SECTION
TAKEAWAY

By listing count, Rio and LA look like the two markets to back. By sellable supply, only LA qualifies — Amsterdam's weakness is a supply cap, not a demand gap, and Rio needs conversion, not acquisition. Three of four allocation calls reverse.